

# **Sales Analytics Dashboard**

**Name: MohammmadHashim Kadivar**

**Domain: Data Analytics & Business Intelligence (Excel + SQL +  
Power BI/ Tableau)**

**Date: 22/05/2026**

## **Task-2: Advanced Data Analysis, SQL Joins & Business KPI Dashboard.**

- **Project Overview:**

This project focuses on analyzing supermarket sales data using SQL, Excel, and Power BI. The objective of the project was to identify business performance trends, customer behavior, profit analysis, and regional sales insights through data analytics and visualization techniques. The project included database creation, SQL JOIN operations, KPI calculations, dashboard development, and business performance analysis.

- **Tools Used:**

| Tools               | Purpose                                |
|---------------------|----------------------------------------|
| Excel               | Data Cleaning, Analysis & Pivot Tables |
| DB Browser (SQLite) | Data Extraction, SQL Queries           |
| Power BI            | Dashboard Visualization                |

- **Business Performance Analysis:**

1. **Which region generates the highest revenue?**

The **West Region** generated the highest revenue among all regions. This indicates strong customer demand and better sales performance in that area. The region contributed significantly to overall business growth and profitability.

2. **Which category has the highest profit margin?**

The **Technology Category** achieved the highest profit margin. Technology products generated better profit compared to Furniture and Office Supplies. This category plays an important role in increasing overall business profit.

3. **Are discounts reducing profitability?**

Yes, higher discounts are reducing profitability in several transactions. Orders with large discount percentages showed lower profit margins. This indicates that discount strategies should be managed carefully to avoid profit loss.

4. **Which customer segment is most valuable?**

The **Consumer Segment** is the most valuable customer segment. It contributed the highest number of orders and generated strong sales revenue. The segment plays a major role in overall business performance.

5. **Is there a seasonal sales trend?**

Yes, the business shows a clear seasonal sales trend. Sales increased during the final months of the year, especially in **November and December**. This suggests higher customer purchasing activity during festive and holiday seasons.

- **Excel Analysis:**

- **Pivot Tables:**

- **Sales By Region:**

| Sales By Region    |                    |
|--------------------|--------------------|
| Row Labels         | Sum of sales       |
| Central            | 345992.9567        |
| East               | 397446.0149        |
| South              | 286087.2943        |
| West               | 478948.2983        |
| <b>Grand Total</b> | <b>1508474.564</b> |

- **Profit By Category & Region:**

| Profit By Category & Region |                    |
|-----------------------------|--------------------|
| Row Labels                  | Sum of profit      |
| <b>Central</b>              | <b>69198.59134</b> |
| Furniture                   | 22315.40814        |
| Office Supplies             | 21131.6874         |
| Technology                  | 25751.4958         |
| <b>East</b>                 | <b>79489.20298</b> |
| Furniture                   | 23865.95338        |
| Office Supplies             | 24489.526          |
| Technology                  | 31133.7236         |
| <b>South</b>                | <b>57217.45886</b> |
| Furniture                   | 15927.79226        |
| Office Supplies             | 18267.5546         |
| Technology                  | 23022.112          |
| <b>West</b>                 | <b>95789.65966</b> |
| Furniture                   | 35756.47646        |
| Office Supplies             | 30105.8112         |
| Technology                  | 29927.372          |
| <b>Grand Total</b>          | <b>301694.9128</b> |

- **Monthly Sales Trend:**

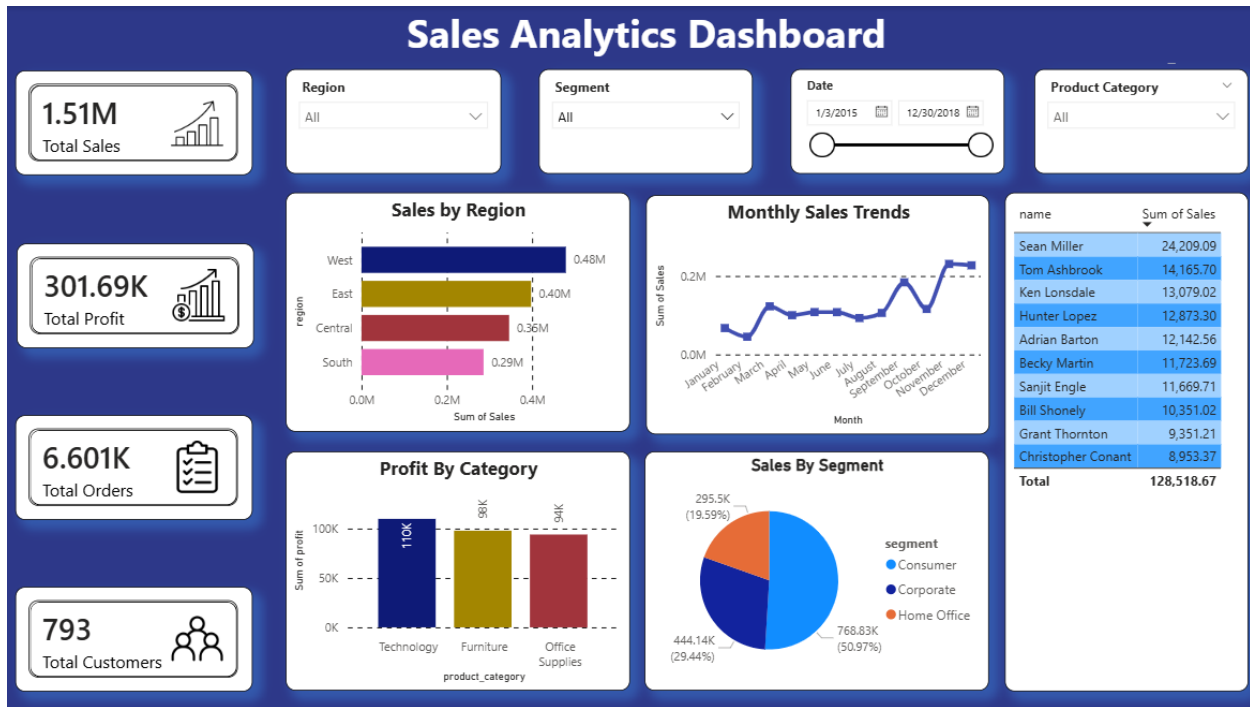
| Monthly Sales Trend |                    |
|---------------------|--------------------|
| Row Labels          | Sum of sales       |
| January             | 110978.366         |
| February            | 71900.7864         |
| March               | 135537.2138        |
| April               | 103268.6731        |
| May                 | 116974.6719        |
| June                | 101263.6485        |
| July                | 103965.7915        |
| August              | 141710.0421        |
| September           | 147032.2641        |
| October             | 119715.022         |
| November            | 180986.3118        |
| December            | 175141.773         |
| <b>Grand Total</b>  | <b>1508474.564</b> |

- **TOP-10 Customers:**

| Top-10 Customers   |                   |
|--------------------|-------------------|
| Row Labels         | Sum of sales      |
| Adrian Barton      | 12142.555         |
| Becky Martin       | 11723.69          |
| Bill Shonely       | 10351.023         |
| Christopher Conant | 8953.368          |
| Grant Thornton     | 9351.212          |
| Hunter Lopez       | 12873.298         |
| Ken Lonsdale       | 13079.015         |
| Sanjit Engle       | 11669.714         |
| Sean Miller        | 24209.094         |
| Tom Ashbrook       | 14165.7           |
| <b>Grand Total</b> | <b>128518.669</b> |

- **Power BI Dashboards:**

**Screenshot:**



[Fig. 1: Sales Analytics Dashboard]



[Fig. 2: Filtered Sales Analytics Dashboard]

- **KPI Overview:**

The dashboard provides important business KPIs including:

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Profit Margin

These KPIs help measure overall business performance and support better decision-making.

- **Regional Performance:**

The regional analysis showed that the West Region generated the highest sales revenue. The East and Central regions also contributed strong sales performance, while some regions showed comparatively lower profitability.

Regional comparison helped identify high-performing business locations and growth opportunities.

- **Category Performance:**

Among all product categories, **Technology** generated the highest profit margin. Furniture contributed strong sales revenue, while Office Supplies maintained stable performance.

Category analysis helped understand which products generate better profitability for the business.

- **Customer Analysis:**

The Consumer segment contributed the highest number of orders and overall revenue. Top customers generated significant sales value, showing the importance of customer retention and loyalty.

Customer analysis also helped identify valuable customer segments for future business strategies.

- **Seasonal sales Trend:**

Sales performance increased during the final months of the year, especially in November and December. This indicates seasonal demand and higher purchasing activity during festive periods.

The monthly sales trend analysis helped identify peak business periods.

- **Conclusion:**

This project successfully demonstrated the use of SQL, Excel, and Power BI for business intelligence and sales analytics. The dashboard and analysis provided meaningful insights into sales performance, profitability, customer behavior, and business trends.

The project enhanced practical skills in:

- SQL JOIN operations
- KPI calculations
- Data visualization
- Dashboard design
- Business analytics